

**Ques. ARE THE CHARGING PROCEDURES FOR LITHIUM-ION AND LITHIUM-POLYMER BATTERIES DIFFERENT? SHOULD THESE BE CHARGED WHEN FULLY DISCHARGED OR WHENEVER POSSIBLE?**

**Ramakanta Mohanta**

**Ans.** The chemistry is basically the same for both kinds of batteries. So charging procedures for one can be used for the other. Trickle charging is not recommended for lithium batteries but float charging can be useful.

Lithium-ion chemistry cannot accept an overcharge without causing damage to the cell, possibly plating out lithium metal and becoming hazardous.

| Depth of discharge | Discharge cycles |
|--------------------|------------------|
| 100%               | 300 - 500        |
| 50%                | 1200 - 1500      |
| 25%                | 2000 - 2500      |
| 10%                | 3750 - 4700      |

Lithium-ion batteries perform best when these are charged before these get completely discharged. Discharge up to 50 per cent is better for the battery's long-term life than, say, discharges by 90 per cent or complete discharges to zero per cent. Fifty per cent discharges provide much better number of cycles-to-usage ratio. A partial discharge reduces stress and prolongs battery life.

The table above will give you an idea when to go for charging, as it compares the number of discharge/charge cycles a lithium-ion battery can deliver at various depths of discharge (DoD) levels before battery capacity drops to 70 per cent over a period of two to three years or

more. Data shows that commencement of charging when the battery is fully discharged will give you fewer discharge cycles.

**Q2. WHAT IS THE DIFFERENCE BETWEEN A10k, B10k AND C10k POTENTIOMETERS?**

**Pamarthi Kanakaraja**

**A2.** Prefixes A, B and C indicate the tapered pattern of resistive material. Taper is the variation of resistance value related to the movement of the wiper. Variation can be linear, logarithmic, anti-logarithmic or semi-logarithmic.

Unfortunately, use of letters is not standardised and depends on the origin (the USA, Japan or Europe). Same markings can have different meanings.

In Europe, A means lin (linear), B log (logarithmic) and C anti-log. Whereas in Japan, A means log, B lin and D anti-log.

The code also varies between different manufacturers. To avoid any confusion, you may find no letter or its meaning on the body of a potentiometer.

**Q3. HOW CAN WE CONVERT AN ORDINARY AC-DC CONVERTER (SMPS ADAPTOR) INTO A CONSTANT-CURRENT LED DRIVER?**

**A. Samiuddin**

**A3.** An SMPS, or switched-mode power supply, transfers power from an AC source (often mains power) to DC loads such as a personal computer or the battery of a mobile phone. An SMPS is lighter and has a higher power-conversion efficiency than a linear power supply.

An SMPS adaptor is a regulated

DC voltage source with specified output voltage and current. Generally, it comes with different voltage outputs (typically 5V, 6V, 9V, 12V and 24V), current outputs, power ratings and input-output connector types and sizes. In fact, some power adaptors have strictly AC voltage outputs, while others may have DC voltage outputs (regulated DC or unregulated DC).

An LED driver circuit provides and regulates power to an LED or string(s) of LEDs. It has power output at a constant level that is matched to the electrical characteristics of the LED(s). Most common sizes for the supply voltage are 2.3V to 5.5V.

A circuit based on LT1932 can drive up to eight white LEDs. Input voltage is from 2.7V to 4.2V.

For further details refer to the application notes of Linear Technology. You may refer the following link: <http://cds.linear.com/docs/en/datasheet/1932f.pdf>

A compatible power adaptor can therefore be used to power the circuit.

**Q4. WHAT IS INDUSTRY 4.0?**

**Satyajit Samal**

**A4.** Industry 4.0, or the fourth industrial revolution, is the current trend of automation and data exchange in manufacturing technologies. The term originates from the strategy formulated by the German government for ten future projects identified by them. The concept promotes computerisation of manufacturing, which includes cyber-physical systems, the Internet of Things and Cloud computing. Industry 4.0 creates what is called a smartfactory.

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